DANIEL ESPIRITU

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EDUCATION

University of California, Irvine — Irvine, CA
Bachelor's of Science in Computer Science & Physics (Double) | *GPA: 4.0* — June 2029
CHC Honors Program, Chancellor's Scholar, Distinguished Scholar

Moorpark College — Moorpark, CA
Associate of Science in Computer Science, Physics, and Mathematics (Transfer) | *GPA: 3.91* — May 2026
Phi Theta Kappa Honors Society, Dean's List (x4), Honors Program, Computer Science & AI Club Vice Presidents

WORK EXPERIENCE

NASA — Eugene, OR
Software Engineering Intern — Jun 2026 - Aug 2026
- Constructed a high-energy particle simulator to create an array of particle impact heatmaps on the surface of the Jovian moon Europa.
- Implemented multiprocessing and optimized C-level compilation down from Python to cut execution time down from 5 months to 1 hour.
- Identified radiolytic hotspots through heatmaps on Europa and presented findings at NASA conference at University of Colorado, Boulder.

Moorpark College — Moorpark, CA
Course Embedded Tutor — Jan 2025 - Jun 2026
- Facilitated study sessions and exam reviews, increasing average student academic performance by 1.5 letter grades.
- Decreased the drop rate for the specific classes tutored from 45% to 20%.
- Collaborated with the Mathematics Department faculty to form both a Calculus and an Algebra curriculum used by hundreds of students.

Math and Science Center Tutor — Jan 2025 - Jun 2026
- Assembled test review packets for each Calculus section, requiring 0 revisions over a record-long 3 months.
- Tutored students in topics such as Algebra, Calculus, Differential Equations, Physics, General Programming, Data Structures, and Discrete Math.
- Attained a 100% recommendation rate from colleagues and faculty partners to tutor any of the college's lower division math subjects.

iD Tech Camps & Academies — Pasadena, CA
Camp Instructor @ Caltech — May 2025 - Aug 2025
- Taught fundamental concepts from General programming, Machine Learning, Robotics, and Game Design to students from ages 7-17.
- Organized a specialized curriculum, which demystified the teaching process and increased student test scores by 20%.
- Boasted a 95% student satisfaction rate from weekly student and parent evaluation forms.

PROJECTS

Einstein's Playground — Aug 2026 - Present
- All-encompassing undergraduate physics simulator and learner in topics covering everything from kinematics to relativity.
- Demonstrated knowledge of complex physical interactions through low-error numerical integration simulations with RK4 and Boris methods.
- Working on integrating an AI-learner to assist with user understanding of complicated mathematical and physical concepts.
- Planning to launch a paid version in waves with a subscription model to serve thousands of undergraduate students at Moorpark College and UCI.
- Languages/Tools used: TypeScript (React, Node, Express), HTML, CSS

High Energy Particle Simulator (HEPS) for NASA's Europa Clipper Satellite — Aug 2026
- Directly contributed to the Europa Clipper satellite project by identifying future research priority areas on Europa's surface.
- Simulated 27 million particles through a multifarious magnetic field with varying launch locations, angles, initial energies, charges, and masses.
- Parallelized execution by employing Concurrent.futures, and optimized compilation using Numba.njit, pre-compiling Python into C code.
- Presented results at 2026 Europa ICONS conference at CU Boulder, earning "Best Presentation" award.
- Languages/Tools used: Python (NumPy, Matplotlib, SciPy, Concurrent.futures, Numba.njit)

QuakeAI (Artificial Intelligence Los Angeles) — July 2025
- Utilized database of all earthquakes $\geq$5.0 magnitude since 1900, and created an XGBoosted Regressor-based machine learning model.
- The model was able to predict within $\pm$0.5 magnitude and $\pm$100km depth with 85% accuracy on a training set of ~10,000 entries.
- 7 out of 10 of the model's features were custom engineered, with most using vector mathematics to calculate distances from a geoJSON data file.
- Presented project at AILA project showcase, earning "Most Innovative Project" and "Best Use of AI" awards.
- Languages/Tools used: Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, XGBoost), Google Colab, Jupyter Notebook

PROFESSIONAL DEVELOPMENT

NASA — Remote
NASA's Proposal Writing and Evaluation Experience (NPWEE) Lead Engineer — Sep 2025 - Dec 2025
- Assumed position of Lead Engineer to facilitate scientific discussions, guiding team in the right direction regarding astrophysical applications.
- Worked alongside international subject matter experts (SMEs) in space science to realistically tailor our team's proposal.
- Conducted research to tackle one of NASA's pain points and design an innovative solution.

Artificial Intelligence Los Angeles (AILA) — Los Angeles, CA
Program Member — May 2025 - Aug 2025
- Individually constructed an earthquake prediction machine learning model, which was able to achieve an 85% prediction accuracy.
- Applied optimization tactics to multiple machine learning projects that decreased training time by an average of 40%.
- Leveraged Matplotlib and NumPy within a Python environment to conduct in-depth error analysis, improving model accuracy by 20%.

SKILLS

Programming Languages: Python, Java, C++, C, C#, SQL, MATLAB, HTML, CSS, JavaScript
ML & Data Science: TensorFlow, Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, SciPy, Numba
Tools & Platforms: Jupyter Notebook, Google Colab, Docker, Git, GitHub, Linux (Arch), Bash